In recent years, many methods have been developed for a widespread of analysis and are marketed as assay kits. The kits used as diagnostic tools range from some simple colorimetric assays to more sophisticated procedures involving antibodies or nucleic acids. The kits include all necessary standards and can be used in manual procedures or using an automated instrument. In biochemical and research laboratories, a biochemist should be able to efficiently handle techniques such as chromatography, centrifugation, electrophoresis and spectrophotometry as they are the major techniques used in biochemistry.

10.1 Chromatography

Mikhail Tsvet, a Russian Botanist invented chromatography in the year 1906, which is derived from the Greek word meaning colour (chroma) and writing (graphein). It is an important technique of identifying, separating and purifying different properties of chemical components. Two main approaches of chromatography are analytical and preparative. Analytical methods uses small amount of sample to identify or characterise them. Preparative methods use larger quantities of samples and are mainly used for isolation process.

10.2 Principle of chromatography

Chromatography is defined as the method of separation of different substances based on their partition coefficients in two immiscible phases. Chromatography involves a stationary and a mobile phase. The sample to be analysed, when placed in the stationary phase (solid, bonded coating) will gradually move along in the same direction as the mobile phase (liquid or gas). A component that is soluble in the stationery phase takes a longer time to travel as compared to the one that is insoluble. Therefore, the differences in their mobility can be ascribed to the interaction of the sample with the mobile and stationary phase. For a compound distributing itself between equal volumes of two immiscible solvents A & B, the distribution or partition co-efficient (Kd) is given as,

$$K_d = \frac{\text{Concentration of a component in the moving phase}}{\text{Concentration of that component in the stationery phase}}$$

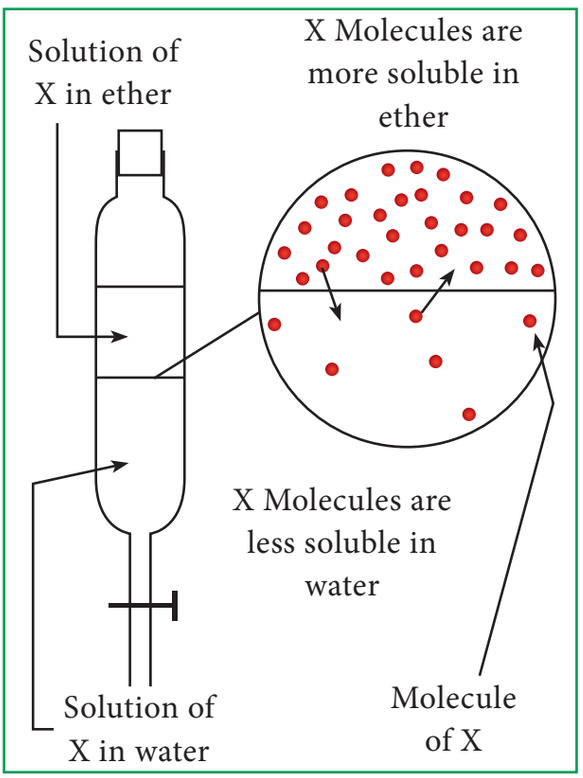

Fig.10.1 : Principle of Chromatography

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